



Operációs rendszerek Bsc

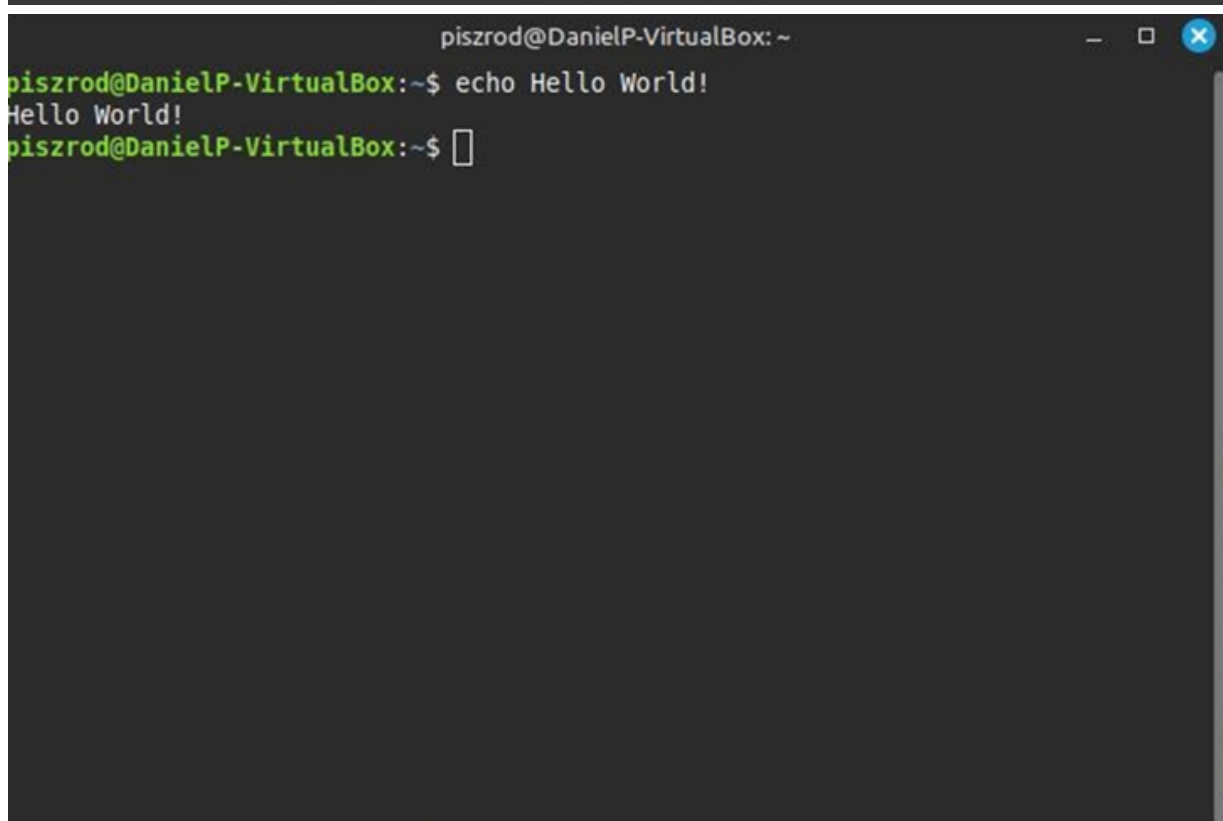
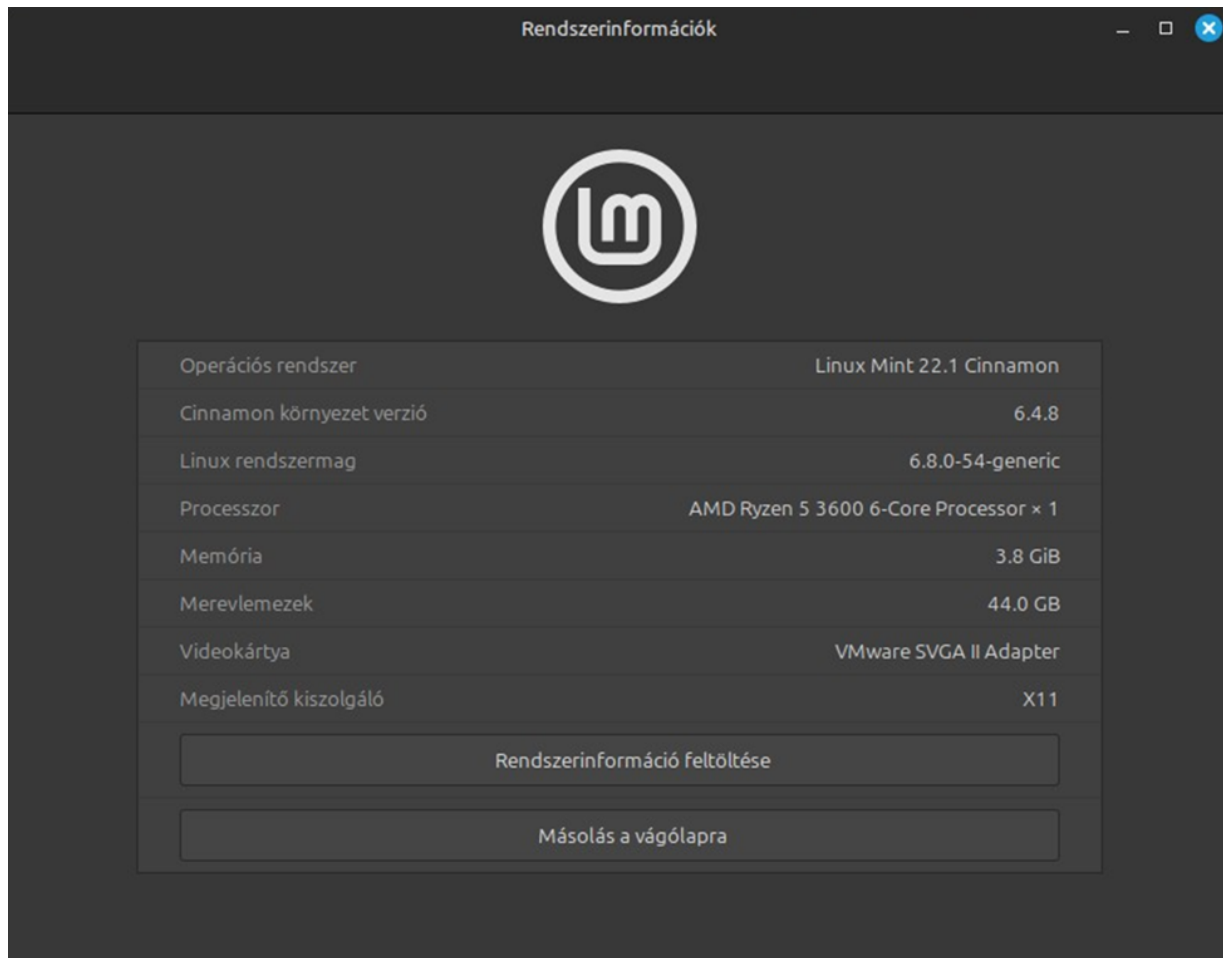
1.Konzultáció

Készítette: Pisztró Dániel

Neptun kód: ELJUVY

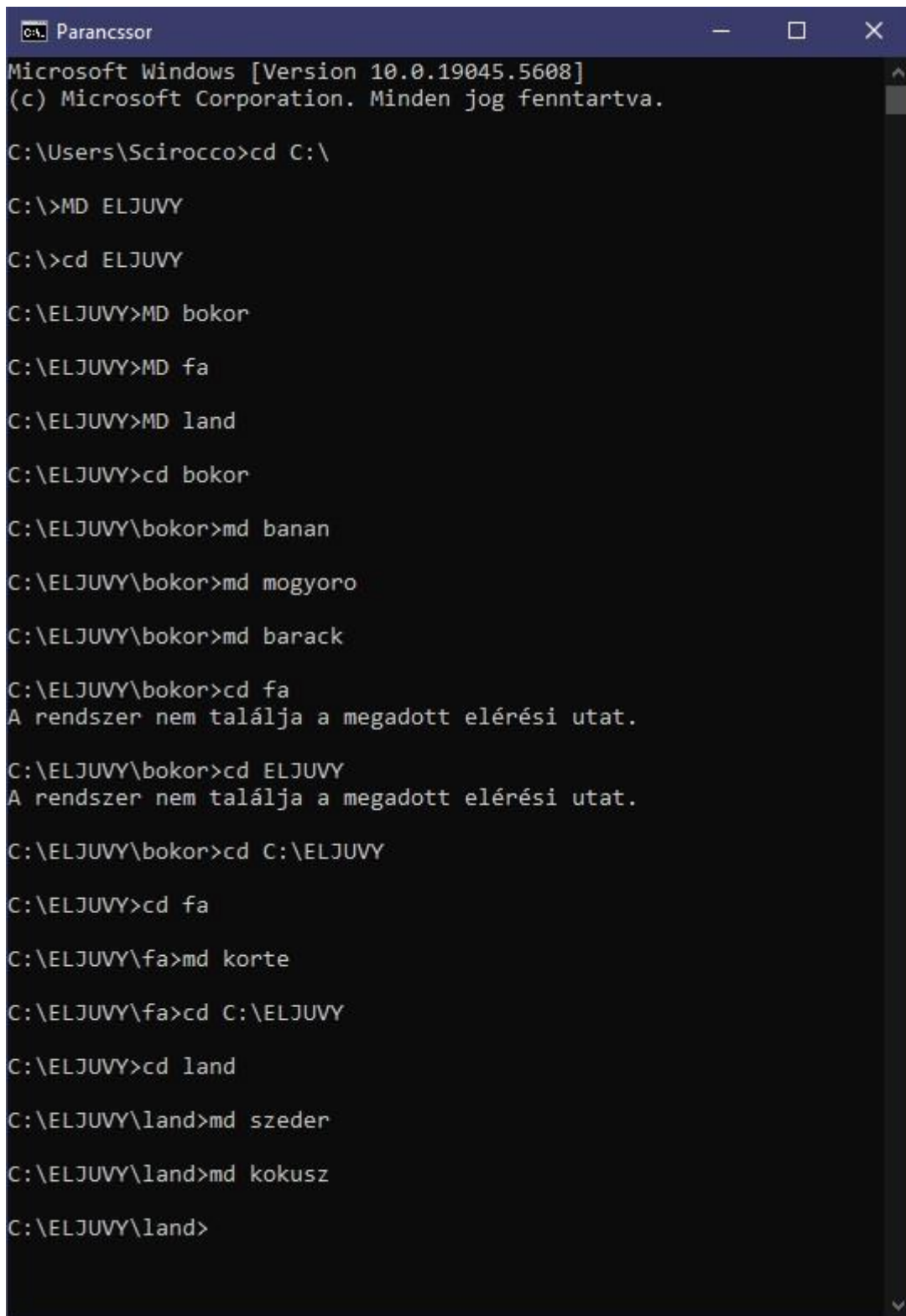
2026.02.27

1. Feladat



2.Feladat

2/a



```
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. Minden jog fenntartva.

C:\Users\Scirocco>cd C:\

C:\>MD ELJUVY

C:\>cd ELJUVY

C:\ELJUVY>MD bokor

C:\ELJUVY>MD fa

C:\ELJUVY>MD land

C:\ELJUVY>cd bokor

C:\ELJUVY\bokor>md banan

C:\ELJUVY\bokor>md mogyoro

C:\ELJUVY\bokor>md barack

C:\ELJUVY\bokor>cd fa
A rendszer nem találja a megadott elérési utat.

C:\ELJUVY\bokor>cd ELJUVY
A rendszer nem találja a megadott elérési utat.

C:\ELJUVY\bokor>cd C:\ELJUVY

C:\ELJUVY>cd fa

C:\ELJUVY\fa>md korte

C:\ELJUVY\fa>cd C:\ELJUVY

C:\ELJUVY>cd land

C:\ELJUVY\land>md szeder

C:\ELJUVY\land>md kokusz

C:\ELJUVY\land>
```

2/b

```
Parancssor
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. Minden jog fenntartva.

C:\Users\Scirocco>xcopy "C:\ELJUVY\land\szeder" "C:\ELJUVY\fa\szeder" /e
Does C:\ELJUVY\fa\szeder specify a file name
or directory name on the target
(F = file, D = directory)? d
0 File(s) copied

C:\Users\Scirocco>xcopy "C:\ELJUVY\bokor\banan" "C:\ELJUVY\fa\banan" /e
Does C:\ELJUVY\fa\banan specify a file name
or directory name on the target
(F = file, D = directory)? d
0 File(s) copied

C:\Users\Scirocco>
```

2/c

```
Parancssor

C:\Users\Scirocco>move "C:\ELJUVY\bokor\barack" "C:\ELJUVY\fa\barack"
1 dir(s) moved.

C:\Users\Scirocco>move "C:\ELJUVY\land\kokusz" "C:\ELJUVY\fa\kokusz"
1 dir(s) moved.

C:\Users\Scirocco>
```

2/d

```
Parancssor

C:\Users\Scirocco>rmdir /s "C:\ELJUVY\land"
C:\ELJUVY\land, Are you sure (Y/N)? y

C:\Users\Scirocco>copy NUL "C:\ELJUVY\bokor\banan\leiras.txt"
1 file(s) copied.

C:\Users\Scirocco>copy NUL "C:\ELJUVY\fa\felsorolas.txt"
1 file(s) copied.

C:\Users\Scirocco>
```

2/e

```
Parancssor

C:\ELJUVY\bokor\banan>copy con leiras.txt
a barack kerek
Overwrite leiras.txt? (Yes/No/All): y
a barack puha
a barack finom^Z
        1 file(s) copied.

C:\ELJUVY\bokor\banan>cd C:\ELJUVY\fa

C:\ELJUVY\fa>copy con felsorolas.txt
Csaba
Lili
Tibor
Vencel
Viola^Z
        1 file(s) copied.

C:\ELJUVY\fa>
```

2/f

```
Parancssor

C:\ELJUVY>tree /f
Folder PATH listing
Volume serial number is 924E-9547
C:..
|_ bokor
|   |_ banan
|       leiras.txt
|   |_ mogyoro
|_ fa
    |_ felsorolas.txt
    |_ banan
    |_ barack
    |_ kokusz
    |_ korte
    |_ szeder

C:\ELJUVY>
```

2/g

```
Parancssor
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. Minden jog fenntartva.

C:\Users\Scirocco>cd C:\ELJUVY

C:\ELJUVY>dir ?e* /s
Volume in drive C has no label.
Volume Serial Number is 924E-9547

Directory of C:\ELJUVY\bokor\banan

2026. 03. 18.  00:47                45 leiras.txt
                1 File(s)                45 bytes

Directory of C:\ELJUVY\fa

2026. 03. 18.  00:50                33 felsorolas.txt
                1 File(s)                33 bytes

Total Files Listed:
                2 File(s)                78 bytes
                0 Dir(s)  374 540 374 016 bytes free

C:\ELJUVY>
```

2/h

```
Parancssor

C:\ELJUVY>attrib -r C:\ELJUVY\fa\felsorolas.txt

C:\ELJUVY>
```

2/i (1/2)

```
Parancssor
C:\ELJUVY>dir /s
Volume in drive C has no label.
Volume Serial Number is 924E-9547

Directory of C:\ELJUVY

2026. 03. 18. 00:24 <DIR>      .
2026. 03. 18. 00:24 <DIR>      ..
2026. 03. 18. 00:11 <DIR>      bokor
2026. 03. 18. 00:36 <DIR>      fa
0 File(s)              0 bytes

Directory of C:\ELJUVY\bokor

2026. 03. 18. 00:11 <DIR>      .
2026. 03. 18. 00:11 <DIR>      ..
2026. 03. 18. 00:34 <DIR>      banan
2026. 03. 17. 23:27 <DIR>      mogyoro
0 File(s)              0 bytes

Directory of C:\ELJUVY\bokor\banan

2026. 03. 18. 00:34 <DIR>      .
2026. 03. 18. 00:34 <DIR>      ..
2026. 03. 18. 00:47      45 leiras.txt
1 File(s)              45 bytes

Directory of C:\ELJUVY\bokor\mogyoro

2026. 03. 17. 23:27 <DIR>      .
2026. 03. 17. 23:27 <DIR>      ..
0 File(s)              0 bytes

Directory of C:\ELJUVY\fa

2026. 03. 18. 00:36 <DIR>      .
2026. 03. 18. 00:36 <DIR>      ..
2026. 03. 17. 23:27 <DIR>      banan
2026. 03. 17. 23:27 <DIR>      barack
2026. 03. 18. 00:50      33 felsorolas.txt
2026. 03. 17. 23:29 <DIR>      kokusz
2026. 03. 17. 23:28 <DIR>      korte
2026. 03. 17. 23:29 <DIR>      szeder
1 File(s)              33 bytes

Directory of C:\ELJUVY\fa\banan
```


2/i (2/2)

```
Kijelölés Parancssor

Directory of C:\ELJUVY\fa\banan
2026. 03. 17. 23:27 <DIR> .
2026. 03. 17. 23:27 <DIR> ..
0 File(s) 0 bytes

Directory of C:\ELJUVY\fa\barack
2026. 03. 17. 23:27 <DIR> .
2026. 03. 17. 23:27 <DIR> ..
0 File(s) 0 bytes

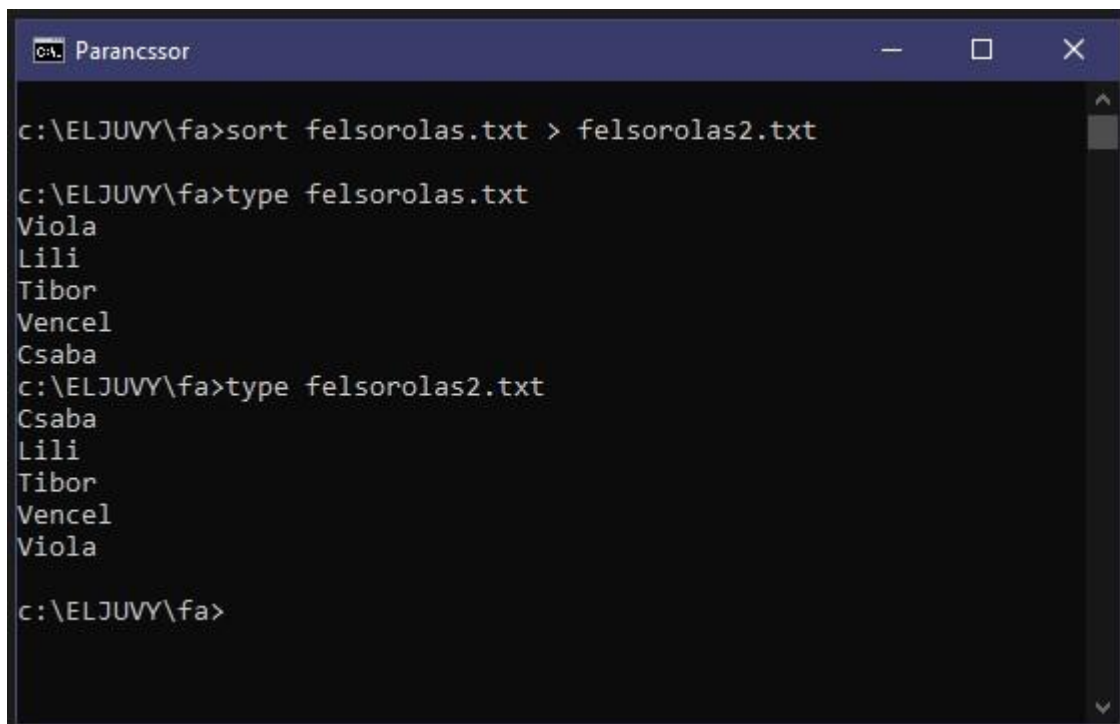
Directory of C:\ELJUVY\fa\kokusz
2026. 03. 17. 23:29 <DIR> .
2026. 03. 17. 23:29 <DIR> ..
0 File(s) 0 bytes

Directory of C:\ELJUVY\fa\korte
2026. 03. 17. 23:28 <DIR> .
2026. 03. 17. 23:28 <DIR> ..
0 File(s) 0 bytes

Directory of C:\ELJUVY\fa\szeder
2026. 03. 17. 23:29 <DIR> .
2026. 03. 17. 23:29 <DIR> ..
0 File(s) 0 bytes

Total Files Listed:
2 File(s) 78 bytes
29 Dir(s) 290 155 794 432 bytes free
```


2/j

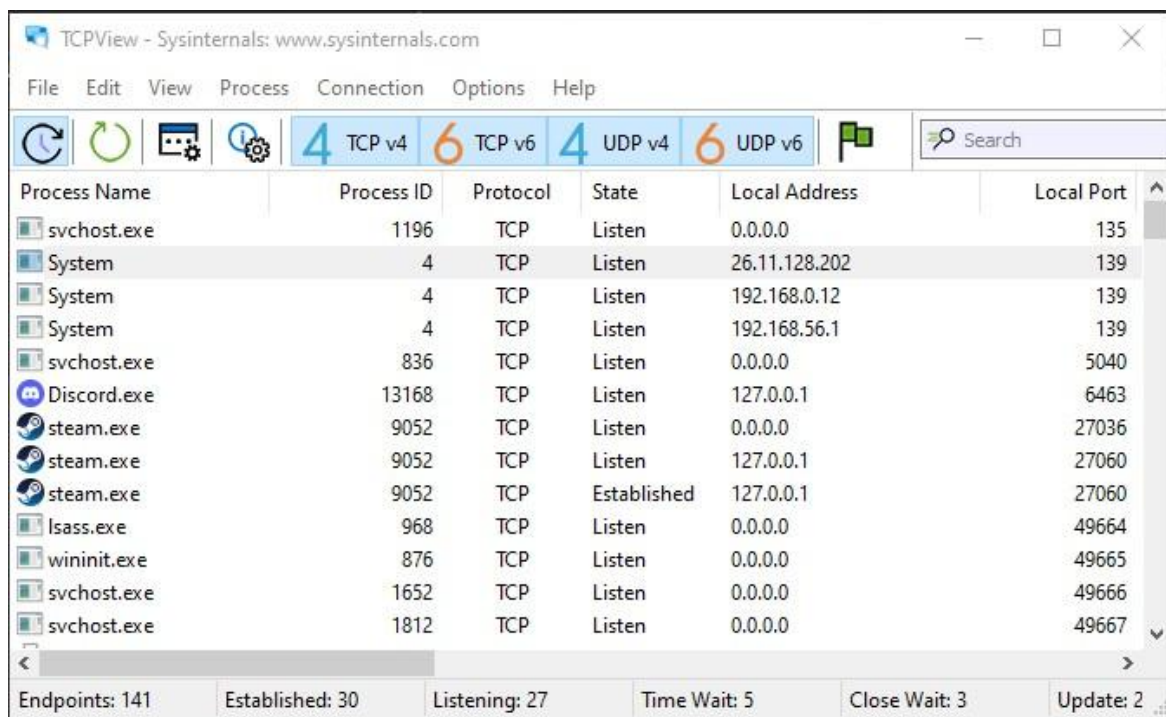


```
c:\ELJUVY\fa>sort felsorolas.txt > felsorolas2.txt

c:\ELJUVY\fa>type felsorolas.txt
Viola
Lili
Tibor
Vencel
Csaba
c:\ELJUVY\fa>type felsorolas2.txt
Csaba
Lili
Tibor
Vencel
Viola

c:\ELJUVY\fa>
```

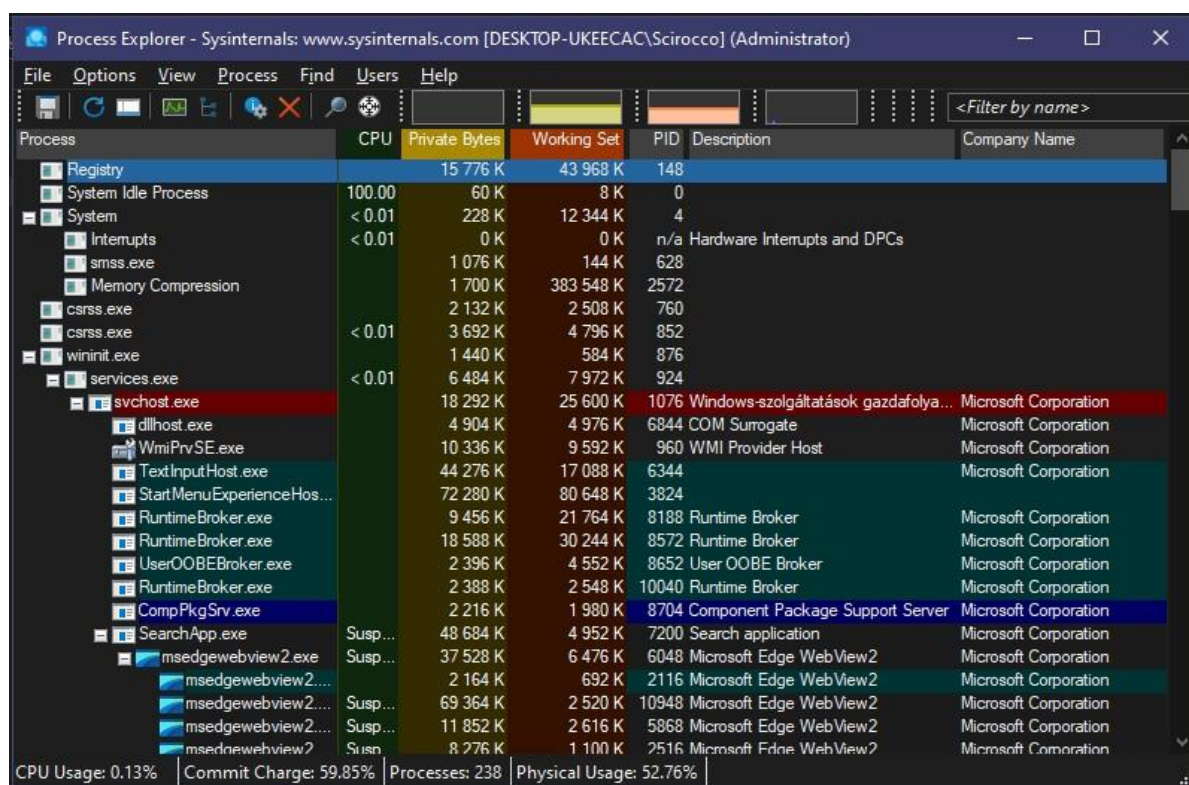
2. Task



Process Name	Process ID	Protocol	State	Local Address	Local Port
svchost.exe	1196	TCP	Listen	0.0.0.0	135
System	4	TCP	Listen	26.11.128.202	139
System	4	TCP	Listen	192.168.0.12	139
System	4	TCP	Listen	192.168.56.1	139
svchost.exe	836	TCP	Listen	0.0.0.0	5040
Discord.exe	13168	TCP	Listen	127.0.0.1	6463
steam.exe	9052	TCP	Listen	0.0.0.0	27036
steam.exe	9052	TCP	Listen	127.0.0.1	27060
steam.exe	9052	TCP	Established	127.0.0.1	27060
lsass.exe	968	TCP	Listen	0.0.0.0	49664
wininit.exe	876	TCP	Listen	0.0.0.0	49665
svchost.exe	1652	TCP	Listen	0.0.0.0	49666
svchost.exe	1812	TCP	Listen	0.0.0.0	49667

Endpoints: 141 Established: 30 Listening: 27 Time Wait: 5 Close Wait: 3 Update: 2

A **TCPView** részletesen megjeleníti a rendszer összes TCP- és UDP-végpontját, beleértve a helyi és távoli címeket, valamint a TCP kapcsolatok állapotát. 4/c (1/3)



Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Registry		15 776 K	43 968 K	148		
System Idle Process	100.00	60 K	8 K	0		
System	< 0.01	228 K	12 344 K	4		
System	< 0.01	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1 076 K	144 K	628		
Memory Compression		1 700 K	383 548 K	2572		
csrss.exe		2 132 K	2 508 K	760		
csrss.exe	< 0.01	3 692 K	4 796 K	852		
wininit.exe		1 440 K	584 K	876		
services.exe	< 0.01	6 484 K	7 972 K	924		
svchost.exe		18 292 K	25 600 K	1076	Windows-szolgáltatások gazdafolya...	Microsoft Corporation
dllhost.exe		4 904 K	4 976 K	6844	COM Surrogate	Microsoft Corporation
WmiPrvSE.exe		10 336 K	9 592 K	960	WMI Provider Host	Microsoft Corporation
TextInputHost.exe		44 276 K	17 088 K	6344		Microsoft Corporation
StartMenuExperienceHos...		72 280 K	80 648 K	3824		
RuntimeBroker.exe		9 456 K	21 764 K	8188	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		18 588 K	30 244 K	8572	Runtime Broker	Microsoft Corporation
UserOOBEBroker.exe		2 396 K	4 552 K	8652	User OOBEBroker	Microsoft Corporation
RuntimeBroker.exe		2 388 K	2 548 K	10040	Runtime Broker	Microsoft Corporation
CompPkgSrv.exe		2 216 K	1 980 K	8704	Component Package Support Server	Microsoft Corporation
SearchApp.exe	Susp...	48 684 K	4 952 K	7200	Search application	Microsoft Corporation
msedgewebview2.exe	Susp...	37 528 K	6 476 K	6048	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	Susp...	2 164 K	692 K	2116	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	Susp...	69 364 K	2 520 K	10948	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	Susp...	11 852 K	2 616 K	5868	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	Susp...	8 276 K	1 100 K	2516	Microsoft Edge WebView2	Microsoft Corporation

CPU Usage: 0.13% Commit Charge: 59.85% Processes: 238 Physical Usage: 52.76%

A **Process Explorer** információkat jelenít meg arról, hogy mely leírók és DLL-folyamatok lettek megnyitva és betöltve.

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

Time ... Process Name PID Operation Path Result Detail

17:21:...	lsass.exe	968	ReadFile	C:\Windows\System32\lsasrv.dll	SUCCESS	Offset: 1 632 768, Length: 13 ...
17:21:...	svchost.exe	2812	ReadFile	C:\Windows\System32\Windows.State...	SUCCESS	Offset: 5 473 792, Length: 16 ...
17:21:...	Explorer.EXE	6480	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Name
17:21:...	Explorer.EXE	6480	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: HandleTags, HandleT...
17:21:...	Explorer.EXE	6480	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: HandleTags, HandleT...
17:21:...	Explorer.EXE	6480	RegOpenKey	HKCU\Software\Classes\Applications\...	NAME NOT FOUND	Desired Access: Read
17:21:...	Explorer.EXE	6480	RegOpenKey	HKCR\Applications\Procmon64.exe	NAME NOT FOUND	Desired Access: Read
17:21:...	Explorer.EXE	6480	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Name
17:21:...	Explorer.EXE	6480	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: HandleTags, HandleT...
17:21:...	Explorer.EXE	6480	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Name
17:21:...	Explorer.EXE	6480	RegOpenKey	HKCU\Software\Classes\Applications\...	NAME NOT FOUND	Desired Access: Read
17:21:...	Explorer.EXE	6480	RegOpenKey	HKCR\Applications\Procmon64.exe	NAME NOT FOUND	Desired Access: Read
17:21:...	Explorer.EXE	6480	CreateFile	C:\Users\Scirocco\Desktop\Suli\2.félé...	SUCCESS	Desired Access: Read Attribut...
17:21:...	Explorer.EXE	6480	QueryBasicInfor...	C:\Users\Scirocco\Desktop\Suli\2.félé...	SUCCESS	CreationTime: 2024. 06. 20. 2...
17:21:...	Explorer.EXE	6480	CloseFile	C:\Users\Scirocco\Desktop\Suli\2.félé...	SUCCESS	
17:21:...	Explorer.EXE	6480	RegQueryKey	HKLM	SUCCESS	Query: HandleTags, HandleT...
17:21:...	Explorer.EXE	6480	RegOpenKey	HKLM\Software\Microsoft\Windows\C...	SUCCESS	Desired Access: Query Value
17:21:...	Explorer.EXE	6480	RegQueryValue	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_DWORD, Length...
17:21:...	Explorer.EXE	6480	RegCloseKey	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	
17:21:...	ctfmon.exe	1724	ReadFile	C:\Windows\System32\InputService.dll	SUCCESS	Offset: 4 537 344, Length: 16 ...
17:21:...	Explorer.EXE	6480	RegQueryKey	HKCU	SUCCESS	Query: HandleTags, HandleT...
17:21:...	Explorer.EXE	6480	RegOpenKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Desired Access: Query Value
17:21:...	Explorer.EXE	6480	RegQueryValue	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_DWORD, Length...
17:21:...	Explorer.EXE	6480	RegCloseKey	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	
17:21:...	Explorer.EXE	6480	QueryStandardI...	C:\Users\Scirocco\AppData\Local\Mic...	SUCCESS	AllocationSize: 933 888, End...

Showing 690 294 of 1 286 199 events (53%) | Backed by virtual memory

A **ProcessMonitor** egy monitorozási eszköz, amely valós idejű fájlrendszert, beállításjegyzéket és folyamat-/száltevékenységet jelenít meg.

Autoruns - Sysinternals: www.sysinternals.com (Administrator) [DESKTOP-UKEECAC\Scirocco]

File Search Entry User Options Category Help

Quick Filter

Codecs Boot Execute Image Hijacks Applnit Known DLLs Winlogon

Winsock Providers Print Monitors LSA Providers Network Providers WMI Office

Everything Logon Explorer Internet Explorer Scheduled Tasks Services Drivers

Autoruns Entry	Description	Publisher
Logon		
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run		
☒ CCXProcess	CCXProcess	(Verified) Adobe Inc.
☒ Discord	Update	(Verified) Discord Inc.
☒ EpicGamesLauncher	EpicGamesLauncher	(Verified) Epic Games Inc.
☒ MicrosoftEdgeAutoLaunch_B0CCBA2A956957B86FF47BE94D1A...	Microsoft Edge	(Verified) Microsoft Corporation
☒ OneDrive	Microsoft OneDrive	(Verified) Microsoft Corporation
☒ Opera Browser Assistant	Opera Browser Assistant	(Verified) Opera Norway AS
☒ Steam	Steam	(Verified) Valve Corp.
☒ Wargaming.net Game Center	Wargaming.net Game Center	(Verified) Wargaming Group Limit...
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce		
☒ Application Restart #0	Microsoft Visual Studio 2022	(Verified) Microsoft Corporation

Ready

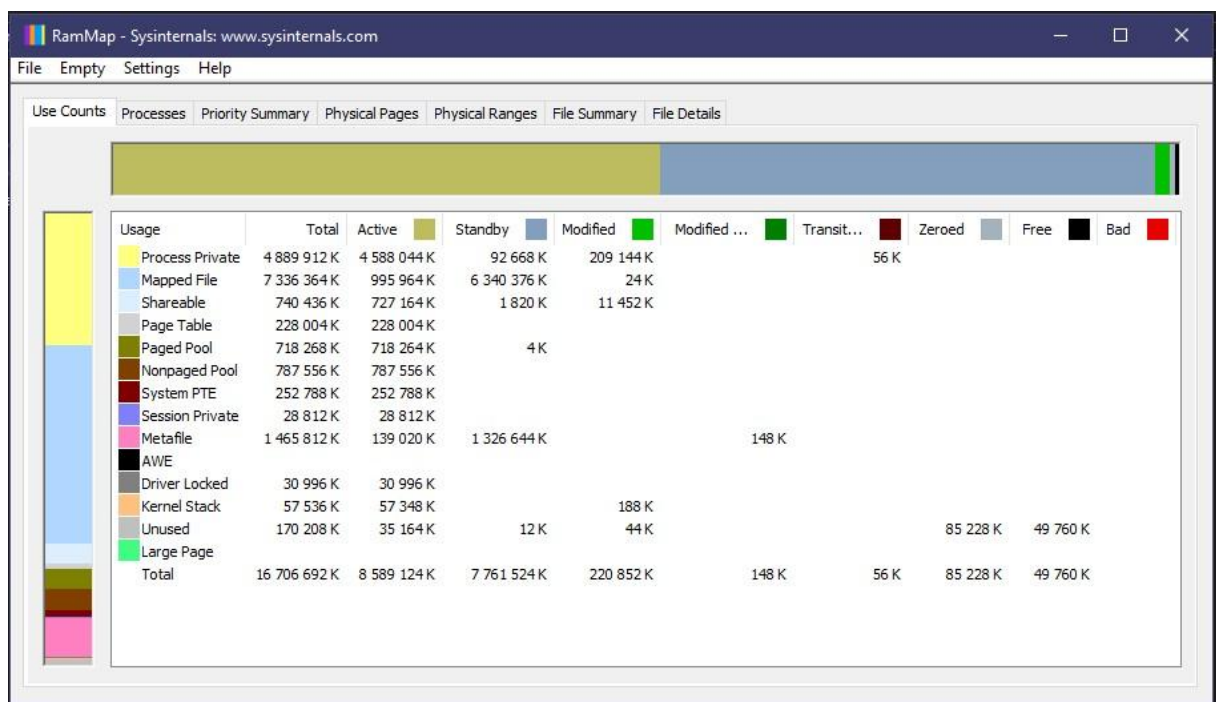
Az **Autoruns** megmutatja, hogy milyen programok futnak a rendszerindítás vagy bejelentkezés során.


```
Administrator: Parancssor
c:\logonSessions>logonsessions64 -p

LogonSessions v1.41 - Lists logon session information
Copyright (C) 2004-2020 Mark Russinovich
Sysinternals - www.sysinternals.com

[0] Logon session 00000000:000003e7:
  User name:      WORKGROUP\DESKTOP-UKEECAC$
  Auth package:   NTLM
  Logon type:     (none)
  Session:        0
  Sid:            S-1-5-18
  Logon time:     2026. 03. 17. 11:04:25
  Logon server:
  DNS Domain:
  UPN:
    968: lsass.exe
    100: winlogon.exe
    1076: svchost.exe
    1248: svchost.exe
    1364: svchost.exe
```

A **LogonSessions** felsorolja az aktuális aktív bejelentkezési munkameneteket és a **-p** beállítással az egyes munkamenetekben futó folyamatokat.



A **RAMMap** egy fizikai memóriahasználat elemző segédprogram. Támogatja a memória pillanatképeinek mentését és betöltését.

3.Task

1.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ mkdir -p ELJUVY/bush/banan  
piszrod@DanielP-VirtualBox:~$ mkdir -p ELJUVY/bush/mogyro  
piszrod@DanielP-VirtualBox:~$ mkdir -p ELJUVY/bush/barack  
piszrod@DanielP-VirtualBox:~$ mkdir -p ELJUVY/tree/korte  
piszrod@DanielP-VirtualBox:~$ mkdir -p ELJUVY/land/szeder  
piszrod@DanielP-VirtualBox:~$ mkdir -p ELJUVY/land/kokusz  
piszrod@DanielP-VirtualBox:~$ tree ELJUVY  
ELJUVY  
├── bush  
│   ├── banan  
│   ├── barack  
│   └── mogyro  
├── land  
│   ├── kokusz  
│   └── szeder  
└── tree  
    └── korte  
  
10 directories, 0 files  
piszrod@DanielP-VirtualBox:~$
```

Az **mkdir** parancs létrehozza a kért mappát, a **-p** opció pedig a teljes útvonal létrehozásához szükséges (ha az még nem létezne). A **tree** parancs kilistázza a kért mappát, almappákkal együtt.

2.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ cp -r ELJUVY/land/szeder ELJUVY/tree  
piszrod@DanielP-VirtualBox:~$ cp -r ELJUVY/bush/banan ELJUVY/tree  
piszrod@DanielP-VirtualBox:~$ tree ELJUVY  
ELJUVY  
├── bush  
│   ├── banan  
│   ├── barack  
│   └── mogyro  
├── land  
│   ├── kokusz  
│   └── szeder  
└── tree  
    ├── banan  
    ├── korte  
    └── szeder  
  
12 directories, 0 files  
piszrod@DanielP-VirtualBox:~$
```

3.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ mv ELJUVY/bush/barack ELJUVY/tree  
piszrod@DanielP-VirtualBox:~$ mv ELJUVY/land/kokusz ELJUVY/tree  
piszrod@DanielP-VirtualBox:~$ tree ELJUVY  
ELJUVY  
├── bush  
│   ├── banan  
│   └── mogyro  
├── land  
│   └── szeder  
└── tree  
    ├── banan  
    ├── barack  
    ├── kokusz  
    ├── korte  
    └── szeder  
  
12 directories, 0 files  
piszrod@DanielP-VirtualBox:~$
```

4.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ rm -r ELJUVY/land  
piszrod@DanielP-VirtualBox:~$ touch ELJUVY/bush/banan/description.txt  
piszrod@DanielP-VirtualBox:~$ touch ELJUVY/tree/listing.txt  
piszrod@DanielP-VirtualBox:~$ tree ELJUVY  
ELJUVY  
├── bush  
│   ├── banan  
│   │   └── description.txt  
│   └── mogyro  
└── tree  
    ├── banan  
    ├── barack  
    ├── kokusz  
    ├── korte  
    ├── listing.txt  
    └── szeder  
  
10 directories, 2 files  
piszrod@DanielP-VirtualBox:~$
```

5.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ echo "A málna édes" >> ELJUVY/bush/banan/description.txt  
piszrod@DanielP-VirtualBox:~$ echo "A málna kicsi" >> ELJUVY/bush/banan/description.txt  
piszrod@DanielP-VirtualBox:~$ echo "A málna finom" >> ELJUVY/bush/banan/description.txt  
piszrod@DanielP-VirtualBox:~$ cat ELJUVY/bush/banan/description.txt  
A málna édes  
A málna kicsi  
A málna finom  
piszrod@DanielP-VirtualBox:~$ echo "barack" >> ELJUVY/tree/listing.txt  
piszrod@DanielP-VirtualBox:~$ echo "banán" >> ELJUVY/tree/listing.txt  
piszrod@DanielP-VirtualBox:~$ echo "citrom" >> ELJUVY/tree/listing.txt  
piszrod@DanielP-VirtualBox:~$ echo "alma" >> ELJUVY/tree/listing.txt  
piszrod@DanielP-VirtualBox:~$ echo "szilva" >> ELJUVY/tree/listing.txt  
piszrod@DanielP-VirtualBox:~$ cat ELJUVY/tree/listing.txt  
barack  
banán  
citrom  
alma  
szilva  
piszrod@DanielP-VirtualBox:~$
```

6.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ tree ELJUVY  
ELJUVY  
├── bush  
│   ├── banan  
│   │   └── description.txt  
│   └── mogyro  
└── tree  
    ├── banan  
    ├── barack  
    ├── kokusz  
    ├── korte  
    ├── listing.txt  
    └── szeder  
  
10 directories, 2 files  
piszrod@DanielP-VirtualBox:~$ pwd  
/home/piszrod  
piszrod@DanielP-VirtualBox:~$
```


7.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ find . -type f -name '?e*'  
./config/cinnamon/spices/network@cinnamon.org/network@cinnamon.org.json  
./config/nemo/desktop-metadata  
./config/caja/desktop-metadata  
./local/state/wireplumber/restore-stream  
./local/share/applications/webapp-OnlineChat4519.desktop  
./local/share/ice/firefox/OnlineChat4519/permissions.sqlite  
./local/share/session_migration-cinnamon  
./local/share/recently-used.xbel  
./mozilla/firefox/ax06acna.default-release/serviceworker.txt  
./mozilla/firefox/ax06acna.default-release/search.json.mozlz4  
./mozilla/firefox/ax06acna.default-release/permissions.sqlite  
./mozilla/firefox/ax06acna.default-release/cert9.db  
./mozilla/firefox/ax06acna.default-release/sessionCheckpoints.json  
./mozilla/firefox/ax06acna.default-release/key4.db  
./mozilla/firefox/ax06acna.default-release/webappsstore.sqlite  
./mozilla/firefox/ax06acna.default-release/datareporting/session-state.json  
./mozilla/firefox/ax06acna.default-release/webappsstore.sqlite-wal  
./mozilla/firefox/ax06acna.default-release/sessionstore.jsonlz4  
./cache/mesa_shader_cache/24/4eb5492dc3c16fb77e81580c672642bbad99f1  
./cache/mesa_shader_cache/3b/ee07d00c69337cdc85245500ed8857d140f220  
./cache/mesa_shader_cache/05/0ec4bb3531e83983fe2d0e0e6c40360971d049  
./cache/mesa_shader_cache/a5/4eec25c443e026f17ff8391dea60aec5f6c11d  
./cache/cinnamon/spices/theme/Serena.png  
./cache/cinnamon/spices/theme/Pepper.png  
./cache/cinnamon/spices/theme/Metallic_Cyan.png
```

8.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ chmod a+r ELJUVY/tree/listing.txt  
piszrod@DanielP-VirtualBox:~$
```

9.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ du -sh ELJUVY  
48K    ELJUVY  
piszrod@DanielP-VirtualBox:~$
```

10.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ sort ELJUVY/tree/listing.txt  
alma  
banán  
barack  
citrom  
szilva  
piszrod@DanielP-VirtualBox:~$
```

11.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ wc -w ELJUVY/bush/banan/description.txt  
9 ELJUVY/bush/banan/description.txt  
piszrod@DanielP-VirtualBox:~$
```

4.Task

(A CPU és a memória adatai)

▼ Processor

Processor #0 ▶

Name	AMD Ryzen 5 3600			Cores	1
Specification	AMD Ryzen 5 3600 6-Core Processor			Threads	1
Package	Microcode	0x8701021	Bogomips	7186.5	
Family	23 (17h)	Model	113 (71h)	Stepping	0 (0h)
Flags	MMX, (+), SSE(1, 2, 3, 3S, 4.1, 4.2, 4A), AVX(1, 2), AES, CLMUL, RdRand, Si				More...
Bugs	Fxsave Leak, Null Seg, Retbleed, Smt Rsb, Spectre V1, Spectre V2, Srsos, Si				More...

▼ Cache

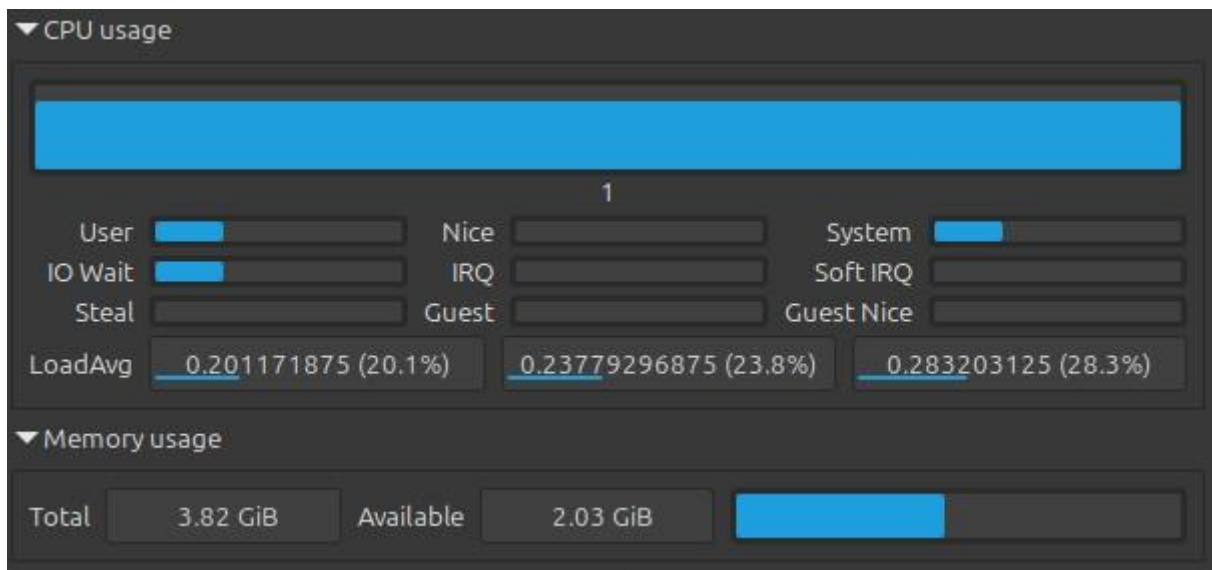
L1 Data	1 x 32 KiB (32 KiB)	16-way	32 sets
L1 Inst.	1 x 32 KiB (32 KiB)	16-way	32 sets
Level 2	1 x 512 KiB (512 KiB)	8-way	1024 sets
Level 3	1 x 32 MiB (32 MiB)	1024-way	512 sets

▼ Motherboard

Vendor		Oracle Corporation		BIOS	
Model		VirtualBox		Vendor innotek GmbH	
Revision		1.2		Version VirtualBox	
				Date 12/01/2006	

▶ Memory

(Az aktuális értékeket jeleníti meg)



(Hardver információkat ír le. Amper és volt számokat, valamint hőfokot)

Hardware Monitor			
Core	Current	Min	Max

a. **top** Processz aktivitást mutató parancs, mely valós idejű módban készít jelentést a futó rendszerről

```

piszrod@DanielP-VirtualBox: ~
top - 19:10:21 up 4 min, 1 user, load average: 0,17, 0,45, 0,24
Tasks: 167 total, 1 running, 166 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0,3 us, 0,0 sy, 0,0 ni, 99,0 id, 0,7 wa, 0,0 hi, 0,0 si, 0,0 st
MiB Mem : 3916,4 total, 2568,2 free, 933,2 used, 646,6 buff/cache
MiB Swap: 3898,0 total, 3898,0 free, 0,0 used. 2983,2 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
 1427 piszrod   20   0 3304692 235728 145748 S   0,3   5,9   0:03.56 cinnamon
 1702 piszrod   20   0 544912  42836  34560 S   0,3   1,1   0:00.24 gnome-t+
    1 root      20   0  22528  13352   9384 S   0,0   0,3   0:00.99 systemd
    2 root      20   0     0     0     0 S   0,0   0,0   0:00.00 kthreadd
    3 root      20   0     0     0     0 S   0,0   0,0   0:00.00 pool_wor
    4 root      0 -20     0     0     0 I   0,0   0,0   0:00.00 kworker+
    5 root      0 -20     0     0     0 I   0,0   0,0   0:00.00 kworker+
    6 root      0 -20     0     0     0 I   0,0   0,0   0:00.00 kworker+
    7 root      0 -20     0     0     0 I   0,0   0,0   0:00.00 kworker+
    8 root      20   0     0     0     0 I   0,0   0,0   0:00.00 kworker+
    9 root      20   0     0     0     0 I   0,0   0,0   0:00.02 kworker+
   10 root      0 -20     0     0     0 I   0,0   0,0   0:00.07 kworker+
   11 root      20   0     0     0     0 I   0,0   0,0   0:00.18 kworker+
   12 root      0 -20     0     0     0 I   0,0   0,0   0:00.00 kworker+
   13 root      20   0     0     0     0 I   0,0   0,0   0:00.00 rcu_tas+
   14 root      20   0     0     0     0 I   0,0   0,0   0:00.00 rcu_tas+
   15 root      20   0     0     0     0 I   0,0   0,0   0:00.00 rcu_tas+

```

b. **vmstat** Ez a parancs a rendszer aktivitásról, a hardverről és rendszerről nyújt információkat.

```

piszrod@DanielP-VirtualBox: ~
piszrod@DanielP-VirtualBox:~$ vmstat -a
procs -----memory----- --swap-- -----io----- -system-- -----cpu-----
 r b  swpd  free  inact active si  so   bi   bo  in  cs us sy id wa st gu
 1 0    0 2628092 145944 990792  0  0  453  42 1330  1 2 1 95 3 0 0
piszrod@DanielP-VirtualBox:~$

```

```

piszrod@DanielP-VirtualBox: ~
piszrod@DanielP-VirtualBox:~$ sudo vmstat -m
[sudo] piszrod jelszava:
Cache               Num    Total    Size    Pages
Acpi-ParseExt         39        39      104       39
Acpi-State          1534      1581       80       51
anon_vma             7387     7488      104       39
anon_vma_chain     13170    13376       64       64
apparmor_auditcache    0         0      240       17
backing_aio           0         0      128       32
bdev_cache           19        19     1664       19
bio-136               42        42      192       21
bio-200              112       112      256       16
bio-264               48        48      320       12

```


c. **w** Ez a parancs megmutatja, ki van bejelentkezve és épp mit csinál a rendszerben.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ w  
19:30:16 up 24 min, 1 user, load average: 0,02, 0,02, 0,06  
USER      TTY      FROM          LOGIN@   IDLE   JCPU   PCPU   WHAT  
piszrod    -        -             19:06    24:09   0.00s   0.04s  lightdm --session-c  
piszrod@DanielP-VirtualBox:~$
```

d. **uptime** Ez a parancs kiírja, hogy mióta fut a rendszer.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ uptime  
19:33:15 up 27 min, 1 user, load average: 0,00, 0,01, 0,04  
piszrod@DanielP-VirtualBox:~$
```

e. **1/8 ps -e** Az összes processzről készít pillanatfelvételt.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ ps -e  
PID TTY      TIME CMD  
1 ?        00:00:01 systemd  
2 ?        00:00:00 kthreadd  
3 ?        00:00:00 pool_workqueue_release  
4 ?        00:00:00 kworker/R-rcu_g  
5 ?        00:00:00 kworker/R-rcu_p  
6 ?        00:00:00 kworker/R-slub_  
7 ?        00:00:00 kworker/R-netns  
9 ?        00:00:00 kworker/0:1-cgroup_destroy  
10 ?       00:00:00 kworker/0:0H-kblockd  
11 ?       00:00:00 kworker/u2:0-flush-8:0  
12 ?       00:00:00 kworker/R-mm_pe  
13 ?       00:00:00 rcu_tasks_kthread  
14 ?       00:00:00 rcu_tasks_rude_kthread  
15 ?       00:00:00 rcu_tasks_trace_kthread  
16 ?       00:00:00 ksoftirqd/0
```

e. **2/8 ps -AIF** Megjeleníti az egyes processzeknek átadott paramétereket is.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ ps -AIF  
F S UID      PID     PPID    C PRI  NI ADDR SZ WCHAN    RSS PSR STIME TTY      TIME CMD  
4 S root      1         0  0  80   0 - 5632 -          13352  0 19:05 ?      00:00:01 /sbin/init splas  
1 S root      2         0  0  80   0 - 0 -          0  0 19:05 ?      00:00:00 [kthreadd]  
1 S root      3         2  0  80   0 - 0 -          0  0 19:05 ?      00:00:00 [pool_workqueue_  
1 I root      4         2  0  60 -20 - 0 -          0  0 19:05 ?      00:00:00 [kworker/R-rcu_g  
1 I root      5         2  0  60 -20 - 0 -          0  0 19:05 ?      00:00:00 [kworker/R-rcu_p  
1 I root      6         2  0  60 -20 - 0 -          0  0 19:05 ?      00:00:00 [kworker/R-slub_  
1 I root      7         2  0  60 -20 - 0 -          0  0 19:05 ?      00:00:00 [kworker/R-netns  
1 I root      9         2  0  80   0 - 0 -          0  0 19:05 ?      00:00:00 [kworker/0:1-cgr  
1 I root     10         2  0  60 -20 - 0 -          0  0 19:05 ?      00:00:00 [kworker/0:0H-kb  
5 I root     11         2  0  80   0 - 0 -          0  0 19:05 ?      00:00:00 [kworker/u2:0-ev  
1 I root     12         2  0  60 -20 - 0 -          0  0 19:05 ?      00:00:00 [kworker/R-mm_pe  
1 I root     13         2  0  80   0 - 0 -          0  0 19:05 ?      00:00:00 [rcu_tasks_kthre  
1 I root     14         2  0  80   0 - 0 -          0  0 19:05 ?      00:00:00 [rcu_tasks_rude  
1 I root     15         2  0  80   0 - 0 -          0  0 19:05 ?      00:00:00 [rcu_tasks_trace  
1 S root     16         2  0  80   0 - 0 -          0  0 19:05 ?      00:00:00 [ksoftirqd/0]  
1 I root     17         2  0  80   0 - 0 -          0  0 19:05 ?      00:00:00 [rcu_preempt]  
1 S root     18         2  0 -40   - - 0 -          0  0 19:05 ?      00:00:00 [migration/0]  
1 S root     19         2  0  9    - - 0 -          0  0 19:05 ?      00:00:00 [idle_inject/0]  
1 S root     20         2  0  80   0 - 0 -          0  0 19:05 ?      00:00:00 [cpuhp/0]  
5 S root     21         2  0  80   0 - 0 -          0  0 19:05 ?      00:00:00 [kdevtmpfs]  
1 I root     22         2  0  60 -20 - 0 -          0  0 19:05 ?      00:00:00 [kworker/R-inet  
1 I root     23         2  0  80   0 - 0 -          0  0 19:05 ?      00:00:00 [kworker/u2:1-ev  
1 S root     24         2  0  80   0 - 0 -          0  0 19:05 ?      00:00:00 [kauditd]
```


e. 3/8 **ps -AIFH** Megjeleníti a processzek szárait.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ ps -AIFH  
F S UID          PID     PPID    C  PRI  NI ADDR  SZ  WCHAN      RSS  PSR  STIME  TTY          TIME CMD  
1 S root           2         0  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [kthreadd]  
1 S root           3         2  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [pool_workqueu  
1 I root           4         2  0  60  -20 -    0 -      0  0  19:05 ?          00:00:00 [kworker/R-rcu  
1 I root           5         2  0  60  -20 -    0 -      0  0  19:05 ?          00:00:00 [kworker/R-rcu  
1 I root           6         2  0  60  -20 -    0 -      0  0  19:05 ?          00:00:00 [kworker/R-slu  
1 I root           7         2  0  60  -20 -    0 -      0  0  19:05 ?          00:00:00 [kworker/R-net  
1 I root           9         2  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [kworker/0:1-c  
1 I root          10         2  0  60  -20 -    0 -      0  0  19:05 ?          00:00:00 [kworker/0:0H-  
5 I root          11         2  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [kworker/u2:0-  
1 I root          12         2  0  60  -20 -    0 -      0  0  19:05 ?          00:00:00 [kworker/R-mm  
1 I root          13         2  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [rcu_tasks_kth  
1 I root          14         2  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [rcu_tasks_rud  
1 I root          15         2  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [rcu_tasks_tra  
1 S root          16         2  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [ksoftirqd/0]  
1 I root          17         2  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [rcu_preempt]  
1 S root          18         2  0 -40   - -    0 -      0  0  19:05 ?          00:00:00 [migration/0]  
1 S root          19         2  0  9    - -    0 -      0  0  19:05 ?          00:00:00 [idle_inject/0  
1 S root          20         2  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [cpuhp/0]  
5 S root          21         2  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [kdevtmpfs]  
1 I root          22         2  0  60  -20 -    0 -      0  0  19:05 ?          00:00:00 [kworker/R-ine  
1 I root          23         2  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [kworker/u2:1-  
1 S root          24         2  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [kauditd]  
1 S root          25         2  0  80   0 -    0 -      0  0  19:05 ?          00:00:00 [khungtaskd]
```

e. 4/8 **ps ax** A szerver processzeit jeleníti meg.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ ps ax  
PID TTY          STAT TIME COMMAND  
1 ?           Ss    0:01 /sbin/init splash  
2 ?           S      0:00 [kthreadd]  
3 ?           S      0:00 [pool_workqueue_release]  
4 ?           I<    0:00 [kworker/R-rcu_g]  
5 ?           I<    0:00 [kworker/R-rcu_p]  
6 ?           I<    0:00 [kworker/R-slub_  
7 ?           I<    0:00 [kworker/R-netns]  
9 ?           I      0:00 [kworker/0:1-cgroup_destroy]  
10 ?          I<    0:00 [kworker/0:0H-kblockd]  
11 ?          I      0:00 [kworker/u2:0-events_power_efficient]  
12 ?          I<    0:00 [kworker/R-mm_pe]  
13 ?          I      0:00 [rcu_tasks_kthread]  
14 ?          I      0:00 [rcu_tasks_rude_kthread]  
15 ?          I      0:00 [rcu_tasks_trace_kthread]  
16 ?          S      0:00 [ksoftirqd/0]  
17 ?          I      0:00 [rcu_preempt]
```

e. 5/8 **ps axu** Megmutatja milyen processzek futnak a szerveren

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ ps axu  
USER          PID  %CPU  %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND  
root           1  0.0  0.3 22528 13352 ?        Ss   19:05   0:01 /sbin/init splash  
root           2  0.0  0.0      0     0 ?        S    19:05   0:00 [kthreadd]  
root           3  0.0  0.0      0     0 ?        S    19:05   0:00 [pool_workqueue_release]  
root           4  0.0  0.0      0     0 ?        I<   19:05   0:00 [kworker/R-rcu_g]  
root           5  0.0  0.0      0     0 ?        I<   19:05   0:00 [kworker/R-rcu_p]  
root           6  0.0  0.0      0     0 ?        I<   19:05   0:00 [kworker/R-slub_  
root           7  0.0  0.0      0     0 ?        I<   19:05   0:00 [kworker/R-netns]  
root           9  0.0  0.0      0     0 ?        I    19:05   0:00 [kworker/0:1-cgroup_destro  
root          10  0.0  0.0      0     0 ?        I<   19:05   0:00 [kworker/0:0H-kblockd]  
root          11  0.0  0.0      0     0 ?        I    19:05   0:00 [kworker/u2:0-flush-8:0]  
root          12  0.0  0.0      0     0 ?        I<   19:05   0:00 [kworker/R-mm_pe]  
root          13  0.0  0.0      0     0 ?        I    19:05   0:00 [rcu_tasks_kthread]  
root          14  0.0  0.0      0     0 ?        I    19:05   0:00 [rcu_tasks_rude_kthread]  
root          15  0.0  0.0      0     0 ?        I    19:05   0:00 [rcu_tasks_trace_kthread]  
root          16  0.0  0.0      0     0 ?        S    19:05   0:00 [ksoftirqd/0]  
root          17  0.0  0.0      0     0 ?        I    19:05   0:00 [rcu_preempt]
```

e. 6/8 **ps**tree Ez a parancs kiírja a processzfát.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ pstree  
systemd--ModemManager--3*[{ModemManager}]  
--NetworkManager--3*[{NetworkManager}]  
--accounts-daemon--3*[{accounts-daemon}]  
--agetty  
--at-spi2-registr--3*[{at-spi2-registr}]  
--avahi-daemon--avahi-daemon  
--colord--3*[{colord}]  
--cron  
--csd-printer--3*[{csd-printer}]  
--cups-browsed--3*[{cups-browsed}]  
--cupsd--dbus  
--dbus-daemon  
--fwupd--5*[{fwupd}]  
--2*[kerneloops]  
--lightdm--Xorg--{Xorg}  
--lightdm--cinnamon-session--agent--3*[{agent}]  
--applet.py
```

e. 7/8 Egy adott PID nevének lekérdezése, pl.: **ps -p 1286 -o comm=**

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ ps -p 1286 -o comm=  
csd-printer  
piszrod@DanielP-VirtualBox:~$
```

e. 8/8 Az 5 legtöbb CPU memóriát fogyasztó processz kiírása: **ps -auxf | sort -nr -k 3 | head -5 -**

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ ps -auxf | sort -nr -k 3 | head -5 -  
piszrod    1427  1.1  5.9 3299560 239052 ?        Sl   19:06   0:33      | \_ cinna  
mon --replace  
root       818   0.3  2.7 398160 109356 tty7     Rsl+ 19:06   0:09 \_ /usr/lib/xorg/Xor  
g -core :0 -seat seat0 -auth /var/run/lightdm/root/:0 -nolisten tcp vt7 -novtswitch  
piszrod    1822  0.1  1.4 494296 58232 ?        Sl   19:07   0:05 mintreport-tray  
USER       PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND  
systemd+   446   0.0  0.1  91212  7936 ?        Ssl  19:06   0:00 /usr/lib/systemd/syst  
emd-timesyncd  
piszrod@DanielP-VirtualBox:~$
```

f. 1/8 **free -b** Megmutatja a memória méretét byte-ban.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ free -b  
              total          used          free      shared  buff/cache   available  
Mem:        4106682368    948928512    2715213824    19042304    687546368    3157753856  
Swap:        4087345152         0    4087345152  
piszrod@DanielP-VirtualBox:~$
```

f. 2/8 **free -k** Megmutatja a memória méretét kilobyte-ban

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ free -k  
              total          used          free      shared  buff/cache   available  
Mem:         4010432         926648        2651576        18596        671472        3083784  
Swap:         3991548           0         3991548  
piszrod@DanielP-VirtualBox:~$
```

f. 3/8 **free -m** Megmutatja a memória méretét megabyte-ban

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ free -m  
              total          used          free      shared  buff/cache   available  
Mem:           3916           904           2589           18           655           3011  
Swap:           3897             0           3897  
piszrod@DanielP-VirtualBox:~$
```

f. 4/8 **free -g** Megmutatja a memória méretét gigabyte-ban

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ free -g  
              total          used          free      shared  buff/cache   available  
Mem:             3             0             2             0             0             2  
Swap:             3             0             3  
piszrod@DanielP-VirtualBox:~$
```

f. 5/8 **free -t** Megmutatja külön sorban, összesítve a fizikai és swap memóriát.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ free -t  
              total          used          free      shared  buff/cache   available  
Mem:         4010432         926508        2651576        18596        671604        3083924  
Swap:         3991548           0         3991548  
Total:        8001980         926508        6643124  
piszrod@DanielP-VirtualBox:~$
```

f. 6/8 **free -o** Újabb csomagverziókból eltávolítottak. Funkciója, hogy kihagyja a buff/cache kiírását.

f. 7/8 **free -s** Adott időközönként kiírja a memória állapotát.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ free -s 5  
              total          used          free      shared  buff/cache   available  
Mem:         4010432         926356        2651324        18920        672344        3084076  
Swap:         3991548           0         3991548  
  
              total          used          free      shared  buff/cache   available  
Mem:         4010432         926348        2651324        18920        672344        3084084  
Swap:         3991548           0         3991548
```


f. 8/8 **free -V** Kírja a free verziószámát.

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ free -V  
free from procps-ng 4.0.4  
piszrod@DanielP-VirtualBox:~$
```

g. 1/13 **iostat -c**

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ iostat -c  
Linux 6.8.0-54-generic (DanielP-VirtualBox)      2026-03-23      _x86_64_      (1 CPU)  
  
avg-cpu:  %user   %nice %system %iowait  %steal   %idle  
           1,21    0,00    0,44    0,81    0,00   97,54  
  
piszrod@DanielP-VirtualBox:~$
```

g. 2/13 **iostat -d**

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ iostat -d  
Linux 6.8.0-54-generic (DanielP-VirtualBox)      2026-03-23      _x86_64_      (1 CPU)  
  
Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd  
sda                4,04        121,29        24,76         0,00     735697    150161         0  
  
piszrod@DanielP-VirtualBox:~$
```

g. 3/13 **iostat -N**

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ iostat -N  
Linux 6.8.0-54-generic (DanielP-VirtualBox)      2026-03-23      _x86_64_      (1 CPU)  
  
avg-cpu:  %user   %nice %system %iowait  %steal   %idle  
           1,21    0,00    0,44    0,80    0,00   97,55  
  
Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd  
sda                4,00        120,15        24,53         0,00     735697    150197         0  
  
piszrod@DanielP-VirtualBox:~$
```

g. 4/13 **iostat -n**

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ iostat 10 5  
Linux 6.8.0-54-generic (DanielP-VirtualBox)      2026-03-23      _x86_64_      (1 CPU)  
  
avg-cpu:  %user   %nice %system %iowait  %steal   %idle  
           1,20    0,00    0,44    0,79    0,00   97,58  
  
Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd  
sda                3,92        117,68        24,04         0,00     735705    150285         0  
  
avg-cpu:  %user   %nice %system %iowait  %steal   %idle  
           0,61    0,00    0,31    0,41    0,00   98,67  
  
Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd  
sda                1,10         0,00         4,80         0,00         0         48         0  
  
avg-cpu:  %user   %nice %system %iowait  %steal   %idle  
           0,20    0,00    0,10    0,00    0,00   99,69
```

g. 5/13 iostat -h

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ iostat -h  
Linux 6.8.0-54-generic (DanielP-VirtualBox)      2026-03-23      _x86_64_      (1 CPU)  
  
avg-cpu:  %user   %nice %system %iowait  %steal   %idle  
           1,2%    0,0%    0,4%    0,8%    0,0%   97,6%  
  
      tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd Device  
      3,86      115,6k      23,7k         0,0k      718,5M     147,0M      0,0k sda  
  
piszrod@DanielP-VirtualBox:~$
```

g. 6/13 iostat -k

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ iostat -k  
Linux 6.8.0-54-generic (DanielP-VirtualBox)      2026-03-23      _x86_64_      (1 CPU)  
  
avg-cpu:  %user   %nice %system %iowait  %steal   %idle  
           1,17    0,00    0,43    0,76    0,00   97,64  
  
Device      tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd  
sda          3,78      113,14      23,17         0,00     735705     150665         0  
  
piszrod@DanielP-VirtualBox:~$
```

g. 7/13 iostat -t

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ iostat -t  
Linux 6.8.0-54-generic (DanielP-VirtualBox)      2026-03-23      _x86_64_      (1 CPU)  
  
2026-03-23 20:54:59  
avg-cpu:  %user   %nice %system %iowait  %steal   %idle  
           1,17    0,00    0,43    0,76    0,00   97,65  
  
Device      tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd  
sda          3,76      112,44      23,03         0,00     735705     150685         0  
  
piszrod@DanielP-VirtualBox:~$
```

g. 8/13 iostat -V

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ iostat -V  
sysstat verzió: 12.6.1  
(C) Sebastien Godard (sysstat <at> orange.fr)  
piszrod@DanielP-VirtualBox:~$
```


g. 9/13 iostat -x

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ iostat -x  
Linux 6.8.0-54-generic (DanielP-VirtualBox)      2026-03-23      _x86_64_      (1 CPU)  
  
avg-cpu:  %user   %nice %system %iowait  %steal   %idle  
           1,15    0,00    0,42    0,74    0,00   97,68  
  
Device            r/s      kB/s    rrqm/s  %rrqm r_await rareq-sz    w/s      kB/s    wrqm/s  %wrqm  
w_await wareq-sz    d/s      kB/s    drqm/s  %drqm d_await dareq-sz    f/s      f_await  aqu-sz   %util  
sda              3,12    110,09    1,43   31,47    7,42    35,30    0,57    22,59    1,03   64,44  
    5,57    39,66    0,00     0,00    0,00    0,00    0,00    0,14    13,38    0,03    0,90  
  
piszrod@DanielP-VirtualBox:~$
```

g. 10/13 iostat -z

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ iostat -z  
Linux 6.8.0-54-generic (DanielP-VirtualBox)      2026-03-23      _x86_64_      (1 CPU)  
  
avg-cpu:  %user   %nice %system %iowait  %steal   %idle  
           1,20    0,00    0,43    0,74    0,00   97,63  
  
Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd  
sda              3,65    108,90      22,35         0,00    735705    151021         0  
  
piszrod@DanielP-VirtualBox:~$
```

g. 11/13 iostat device[...]

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ iostat device[...]  
Linux 6.8.0-54-generic (DanielP-VirtualBox)      2026-03-23      _x86_64_      (1 CPU)  
  
avg-cpu:  %user   %nice %system %iowait  %steal   %idle  
           1,20    0,00    0,42    0,73    0,00   97,65  
  
Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd  
  
piszrod@DanielP-VirtualBox:~$
```

g. 12/13 iostat -p [device,...]

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ iostat -p [device,...]  
Linux 6.8.0-54-generic (DanielP-VirtualBox)      2026-03-23      _x86_64_      (1 CPU)  
  
avg-cpu:  %user   %nice %system %iowait  %steal   %idle  
           1,19    0,00    0,42    0,72    0,00   97,67  
  
Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd  
  
piszrod@DanielP-VirtualBox:~$
```

g.13/13 iostat interval[count]

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ iostat interval[count]  
Linux 6.8.0-54-generic (DanielP-VirtualBox) 2026-03-23 _x86_64_ (1 CPU)  
  
avg-cpu:  %user  %nice %system %iowait  %steal   %idle  
           1,18    0,00    0,42    0,71    0,00   97,69  
  
Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd  
  
piszrod@DanielP-VirtualBox:~$
```

h. sar -n DEV | more

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ sar -n DEV | more  
Linux 6.8.0-54-generic (DanielP-VirtualBox) 2026-03-23 _x86_64_  
(1 CPU)  
  
21:12:55 LINUX RESTART (1 CPU)  
piszrod@DanielP-VirtualBox:~$
```

i. mpstat

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ mpstat  
Linux 6.8.0-54-generic (DanielP-VirtualBox) 2026-03-23 _x86_64_ (1 CPU)  
  
21:15:39 CPU %usr %nice %sys %iowait %irq %soft %steal %guest %gnice %idle  
21:15:39 all 1,17 0,00 0,41 0,66 0,00 0,00 0,00 0,00 0,00 97,76  
piszrod@DanielP-VirtualBox:~$
```

j. 1/3 pmap -d 10

```
piszrod@DanielP-VirtualBox: ~  
piszrod@DanielP-VirtualBox:~$ pmap -d 10  
10: [kworker/0:0H-kblockd]  
Address Kbytes Mode Offset Device Mapping  
mapped: 0K writeable/private: 0K shared: 0K  
piszrod@DanielP-VirtualBox:~$
```